STS 532: Mathematical Statistics II

Homework I

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$\boxed{\textbf{C\&B 7.1}}$ One observation is taken on a discrete random variable X with pmf $f(x|\theta)$ (shown in the table below), where $\theta \in \{1, 2, 3\}$. Find the MLE of θ.

x	$f(x\|1)$	$f(x\|2)$	$f(x\|3)$
0	$\frac{1}{3}$	$\frac{1}{4}$	0
1	$\frac{1}{3}$	$\frac{1}{4}$	0
2	0	$\frac{1}{4}$	$\frac{1}{4}$
3	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{2}$
4	$\frac{1}{6}$	0	$\frac{1}{4}$

$\boxed{\textbf{Solution}}$ Let x be a single observation taken on the random variable X. The likelihood function of this sample is

$$L = f(x|\theta).$$

The MLE (call it $\hat{\theta}$) for θ is the value that maximizes L. When $x = 0$, L is maximized for $\theta = 1$. Following this example, we can write

$$\hat{\theta} = \begin{cases} 1, & \text{if } x = 0 \text{ or } x = 1 \\ 2, & \text{if } x = 2 \\ 3, & \text{if } x = 3 \text{ or } x = 4. \end{cases}$$

When $x = 2$, L is maximized for $\theta = 2$ and $\theta = 3$. So another MLE for θ is

$$\hat{\theta}_2 = \begin{cases} 1, & \text{if } x = 0 \text{ or } x = 1 \\ 3, & \text{if } x = 2, x = 3, \text{ or } x = 4. \end{cases}$$

$\boxed{\textbf{C\&B 7.8}}$ One observation, X, is taken from a $n(0, \sigma^2)$ population.

(a) Find an unbiased estimator of σ^2.

(b) Find the MLE of σ.

(c) Discuss how the method of moments estimator of σ might be found.

$\boxed{\textbf{Solution}}$

(a) Note that

$$\begin{aligned} E[X^2] &= E[X^2] - E^2[X] + E^2[X] && (1) \\ &= V[X] + E^2[X] && (2) \\ &= V[X] && (3) \\ &= \sigma^2. && (4) \end{aligned}$$

Thus, X^2 is an unbiased estimator of σ^2.

(b) The density function of the population is

$$f(x|\sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-x}{2\sigma^2}\right) \tag{5}$$

with $\sigma > 0$ and $-\infty < x < \infty$. For a single observation, the log-likelihood function is then

$$\begin{aligned} \mathscr{L}(x|\sigma) &= \ln\left(L(x|\sigma)\right) && (6) \\ &= \ln\left(f(x|\sigma)\right) && (7) \\ &= \ln\left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-x}{2\sigma^2}\right)\right) && (8) \\ &= -\frac{x}{2\sigma^2} - \ln\left(\sqrt{2\pi}\right) - \ln(\sigma). && (9) \end{aligned}$$

The MLE of σ is the number $\hat{\sigma}$ that maximizes $\mathscr{L}$. To find this number, the equation

$$\frac{\partial \mathscr{L}}{\partial \sigma} = 0 \tag{10}$$

is solved for σ. We have

$$\frac{\partial \mathscr{L}}{\partial \sigma} = x\sigma^{-3} - \sigma^{-1}. \tag{11}$$

Setting this equal to zero yields $\sigma^2 = x^2$, or $\sigma = |x|$. Thus, the MLE for σ is $\hat{\sigma} = |x|$.

(c) Let m_1, m_2 be the first two sample moments and let μ_1', μ_2' be the first two population moments. To find the method-of-moments estimator for σ, we solve the system

$$m_1 = \mu_1' \tag{12}$$

$$m_2 = \mu_2' \tag{13}$$

From part (a), we know that $\mu_2' = \sigma^2$. Since $m_2 = x^2$, our estimator is $\hat{\sigma} = |x|$. In this particular case, the method-of-moments estimator for σ is equal to the MLE for σ.

C&B 7.11 Let $X_1, \ldots, X_n$ be iid with pdf

$$f(x|\theta) = \theta x^{\theta-1}, \;\; 0 \leq x \leq 1, \;\; 0 < \theta < \infty.$$

(a) Find the MLE of θ, and show that its variance tends to 0 as n tends to ∞.

(b) Find the method-of-moments estimator of θ.

Solution

(a) The likelihood function is

$$L = \prod_{i=1}^{n} f(x_i|\theta) \tag{14}$$

$$= \theta^n (x_1 x_2 \cdots x_n)^{\theta-1}. \tag{15}$$

The log-likelihood function is

$$\mathscr{L} = \ln(L) \tag{16}$$

$$= n\ln(\theta) + (\theta - 1)\sum_{i=1}^{n} \ln(x_i). \tag{17}$$

To find the value of θ that maximizes $\mathscr{L}$, we solve

$$\frac{\partial \mathscr{L}}{\partial \theta} = 0.$$

We have

$$\frac{\partial \mathscr{L}}{\partial \theta} = \frac{n}{\theta} + \sum_{i=1}^{n} \ln(x_i). \tag{18}$$

Setting this equal to 0 and solving for θ gives us the desired MLE:

$$\hat{\theta} = \frac{n}{-\sum\limits_{i=1}^{n} \ln(x_i)}.$$

The variance of $\hat{\theta}$ is

$$\text{Var}(\hat{\theta}) = \text{Var}\left[\frac{n}{-\sum\limits_{i=1}^{n} \ln(x_i)}\right] \tag{19}$$

$$= n^2 \text{Var}\left[\frac{1}{-\sum\limits_{i=1}^{n} \ln(X_i)}\right]. \tag{20}$$

Now consider the random variable $W = -\ln(X)$. From Theorem 2.1.5 (from Casella's and Berger's textbook), the pdf of W is

$$f_W(w) = \theta \left(e^{-w}\right)^{\theta-1} \left|-e^{-w}\right| = \theta e^{-w\theta} = \frac{1}{1/\theta} e^{-\frac{w}{1/\theta}}, \;\; w > 0.$$

Clearly, this indicates that W is an exponential random variable. The sum of iid exponential random variables is Gamma. Thus, $Y = -\sum_{i=1}^{n} \ln(X_i)$ is Gamma with $\alpha = n$ and $\beta = \frac{1}{\theta}$. The distribution of $\frac{1}{Y}$ is inverse Gamma with $\alpha = n$ and $\beta = \frac{1}{\theta}$. So

$$\mathrm{Var}(\hat{\theta}) = n^2 \mathrm{Var}\left[\frac{1}{Y}\right] \tag{21}$$

$$= n^2 \frac{(1/\theta)^2}{(n-1)^2(n-2)} \tag{22}$$

Since the highest power of n in the numerator of (22) is 2 and the highest power of n in the denominator of (22) is 3, it is clear that $\mathrm{Var}(\hat{\theta})$ tends to 0 as n tends to ∞.

(b) The method-of-moments estimator for θ is found by solving

$$\bar{X} = E[X]$$

for θ. We have

$$E[X] = \int_0^1 x\theta x^{\theta-1}\, dx \tag{23}$$

$$= \int_0^1 \theta x^{\theta}\, dx \tag{24}$$

$$= \theta \left[\frac{x^{\theta+1}}{\theta+1}\right]_0^1 \tag{25}$$

$$= \frac{\theta}{\theta+1} \tag{26}$$

Solving $\bar{X} = \frac{\theta}{\theta+1}$ for θ yields $\theta = \frac{\bar{X}}{1-\bar{X}}$, which is the method-of-moments estimator.

$\boxed{\textbf{C\&B 7.12}}$ Let $X_1, X_2, \ldots, X_n$ be a random sample from a population with pmf

$$P_\theta(X = x) = \theta^x(1-\theta)^{1-x}, \qquad x = 0 \text{ or } x = 1, \qquad 0 < \theta < \frac{1}{2}.$$

(a) Find the method of moments estimator and MLE of θ.

(b) Find the mean squared errors of each of the estimators.

(c) Which estimator is preferred? Justify your choice.

$\boxed{\textbf{Solution}}$

(a) To find the method of moments estimator of θ, equate $\mu_1' = E[X]$ and $m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$. Since $E[X] = \theta$, the estimator is $\hat{\theta}_1 = \bar{X}$.

To find the MLE of θ, first find the log-likelihood function. For this problem, it will be

$$\mathscr{L} = \ln(\theta)\sum_{i=1}^{n} X_i + \ln(1-\theta)\sum_{i=1}^{n}(1 - X_i). \tag{27}$$

Now set $\frac{\partial \mathscr{L}}{\partial \theta} = 0$. This produces

$$\frac{1}{\theta}\sum_{i=1}^{n} X_i - \frac{1}{1-\theta}\sum_{i=1}^{n}(1 - X_i) = 0 \tag{28}$$

or

$$(1-\theta)\sum_{i=1}^{n} X_i = \theta\sum_{i=1}^{n}(1 - X_i) \tag{29}$$

$$= n\theta - \theta\sum_{i=1}^{n} X_i. \tag{30}$$

Equation (30), of course, implies that $\theta = \bar{X}$. However, we must consider that $\bar{X}$ could be outside the valid range for θ, in which case $\mathscr{L}$ would be maximized at $\theta = \frac{1}{2}$. Thus, the MLE for θ is $\hat{\theta} = \min\left\{\bar{X}, \frac{1}{2}\right\}$.

(b) The mean-squared-error of the method-of-moments estimator is

$$E[(\bar{X} - \theta)^2] = E[\bar{X}^2 - 2\theta\bar{X} + \theta^2] \tag{31}$$

$$= E[\bar{X}^2] - 2\theta E[\bar{X}] + \theta^2. \tag{32}$$

It can easily be shown that $E[\bar{X}] = \theta$. As for $E[\bar{X}^2]$, the calculation is a bit more complicated. We have

$$E[\bar{X}^2] = E\left[\left(\frac{X_1 + X_2 + \ldots + X_n)}{n}\right)^2\right] \tag{33}$$

$$= \frac{1}{n^2}E[(X_1 + X_2 + \ldots + X_n)^2] \tag{34}$$

$$= \frac{1}{n^2}E[X_1^2 + 2X_1(X_2 + \ldots + X_n) + (X_2 + \ldots + X_n)^2]. \tag{35}$$

It is easy to show that all of the moments of the X_i are equal to θ. Furthermore, since the X_i are independent, (35) becomes

$$\frac{1}{n^2}E[X_1^2 + 2X_1(X_2 + \ldots + X_n) + (X_2 + \ldots + X_n)^2] = \frac{1}{n^2}\left[\theta + 2\theta^2(n-1) + E[(X_2 + \ldots + X_n)^2]\right]. \tag{36}$$

Following the emerging pattern, we see that

$$E[\bar{X}^2] = \frac{1}{n^2}\left[n\theta + 2\theta^2 \sum_{i=1}^{n-1}(n-i)\right] \tag{37}$$

$$= \frac{1}{n^2}\left[n\theta + 2\theta^2\left(n(n-1) - \frac{n(n-1)}{2}\right)\right] \tag{38}$$

$$= \frac{\theta}{n} + \frac{\theta^2(n-1)}{n}. \tag{39}$$

Thus, equation (32) becomes

$$E[(\bar{X}-\theta)^2] = \frac{\theta}{n} + \frac{\theta^2(n-1)}{n} - 2\theta^2 + \theta^2 \tag{40}$$

$$= \frac{\theta}{n} + \frac{\theta^2(n-1)}{n} - \theta^2 \tag{41}$$

$$= \frac{\theta + \theta^2 n - \theta^2 - \theta^2 n}{n} \tag{42}$$

$$= \frac{\theta(1-\theta)}{n}. \tag{43}$$

Next we calculate the MSE of the MLE. First note that $Y = n\bar{X}$ is a binomial random variable with n trials and probability of success equal to θ. This will be useful when calculating the MSE. Now, the MSE is

$$E\left[\left(\min\left(\bar{X}, \frac{1}{2}\right) - \theta\right)^2\right] = \frac{1}{n^2}E\left[\left(\min\left(n\bar{X}, \frac{n}{2}\right) - \theta\right)^2\right] \tag{44}$$

$$= \frac{1}{n^2}E\left[\left(\min\left(Y, \frac{n}{2}\right) - n\theta\right)^2\right] \tag{45}$$

$$= \frac{1}{n^2}\sum_{y=0}^{n}\left(\min\left(y, \frac{n}{2}\right) - n\theta\right)^2 \binom{n}{y}\theta^y(1-\theta)^{n-y} \tag{46}$$

$$= \frac{1}{n^2}\sum_{y=0}^{\frac{n}{2}}(y-n\theta)^2\binom{n}{y}\theta^y(1-\theta)^{n-y} + \frac{1}{n^2}\sum_{y=\frac{n}{2}+1}^{n}\left(\frac{n}{2} - n\theta\right)^2\binom{n}{y}\theta^y(1-\theta)^{n-y}. \tag{47}$$

(c) To determine which estimator is better, we find the difference between the two. First note that the MSE for the method-of-moments estimator can be written as

$$E[(\bar{X}-\theta)^2] = \frac{1}{n^2}E[(n\bar{X} - n\theta)^2] \tag{48}$$

$$= \frac{1}{n^2}E[(Y-n\theta)^2] \tag{49}$$

$$= \frac{1}{n^2}\sum_{y=0}^{n}(y-n\theta)^2\binom{n}{y}\theta^y(1-\theta)^{n-y}. \tag{50}$$

Now the difference of the MSEs is

$$\frac{1}{n^2}\sum_{y=0}^{n}(y-n\theta)^2\binom{n}{y}\theta^y(1-\theta)^{n-y}-\frac{1}{n^2}\sum_{y=0}^{\frac{n}{2}}(y-n\theta)^2\binom{n}{y}\theta^y(1-\theta)^{n-y}-\frac{1}{n^2}\sum_{y=\frac{n}{2}+1}^{n}\left(\frac{n}{2}-n\theta\right)^2\binom{n}{y}\theta^y(1-\theta)^{n-y} \tag{51}$$

$$=\frac{1}{n^2}\sum_{y=\frac{n}{2}+1}^{n}\left[(y-n\theta)^2-\left(\frac{n}{2}-n\theta\right)^2\right]\binom{n}{y}\theta^y(1-\theta)^{n-y} \tag{52}$$

$$=\frac{1}{n^2}\sum_{y=\frac{n}{2}+1}^{n}\left(y-n\theta+\frac{n}{2}-n\theta\right)\left(y-n\theta-\frac{n}{2}+n\theta\right)\binom{n}{y}\theta^y(1-\theta)^{n-y} \tag{53}$$

$$=\frac{1}{n^2}\sum_{y=\frac{n}{2}+1}^{n}\left(y+\frac{n}{2}-2n\theta\right)\left(y-\frac{n}{2}\right)\binom{n}{y}\theta^y(1-\theta)^{n-y} \tag{54}$$

The smallest that y can be in this sum is $y=\frac{n}{2}+1$. The largest that θ can be is $\frac{1}{2}$. Thus, unless $\theta=0$, the result in (54) is necessarily positive. This implies that the MSE of the method-of-moments estimator is greater than the MSE of the MLE. Therefore, the MLE is the better estimator.

$\boxed{\textbf{C\&B 7.15(a)}}$ Let $X_1, X_2, \ldots, X_n$ be a sample from the inverse Gamma pdf,

$$f(x|\mu,\lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\left(\frac{-\lambda(x-\mu)^2}{2\mu^2 x}\right) \qquad x > 0.$$

Show that the MLEs of μ and λ are

$$\hat{\mu}_n = \bar{X} \text{ and } \hat{\lambda}_n = \frac{n}{\sum\limits_{i=1}^{n}\left(\frac{1}{X_i} - \frac{1}{\bar{X}}\right)}.$$

$\boxed{\textbf{Solution}}$ The log-likelihood function for this sample is

$$\mathscr{L} = \frac{n}{2}\ln\left(\frac{\lambda}{2\pi}\right) + \frac{3}{2}\sum_{i=1}^{n}\ln\left(\frac{1}{X_i}\right) - \frac{\lambda}{2\mu^2}\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{X_i}. \tag{55}$$

Setting $\frac{\partial \mathscr{L}}{\partial \mu}$ equal to zero yields

$$\begin{aligned}
\frac{\lambda}{\mu^3}\left[\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{X_i} + \sum_{i=1}^{n}\frac{\mu(X_i-\mu)}{X_i}\right] &= \frac{\lambda}{\mu^3}\sum_{i=1}^{n}\frac{X_i^2 - 2X_i\mu + \mu^2 + X_i\mu - \mu^2}{X_i} && (56)\\
&= \frac{\lambda}{\mu^3}\sum_{i=1}^{n}\frac{X_i^2 - X_i\mu}{X_i} && (57)\\
&= \frac{\lambda}{\mu^3}\sum_{i=1}^{n}(X_i - \mu) && (58)\\
&= 0. && (59)
\end{aligned}$$

Therefore, $\sum\limits_{i=1}^{n}\mu = \sum\limits_{i=1}^{n}X_i$; i.e., $\mu = \bar{X}$. Thus, the MLE for μ is $\hat{\mu}_n = \bar{X}$. To find the MLE for λ we begin by setting $\frac{\partial \mathscr{L}}{\partial \lambda} = 0$. After a few simple algebraic manipulations, this yields

$$\lambda = \frac{n}{\frac{1}{\mu^2}\sum\limits_{i=1}^{n}\frac{(X_i-\mu)^2}{X_i}}. \tag{60}$$

The denominator of the above can be written as

$$\begin{aligned}
\frac{1}{\mu^2}\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{X_i} &= \frac{1}{\mu^2}\sum_{i=1}^{n}\frac{X_i^2 - 2X_i\mu + \mu^2}{X_i} && (61)\\
&= \sum_{i=1}^{n}\frac{X_i}{\mu^2} - \sum_{i=1}^{n}\frac{2}{\mu} + \sum_{i=1}^{n}\frac{1}{X_i}. && (62)
\end{aligned}$$

Substituting $\hat{\mu}_n$ for μ in the above equation makes the right-hand side become

$$\begin{aligned}
\sum_{i=1}^{n}\frac{X_i}{\bar{X}^2} - \bar{X}\sum_{i=1}^{n}\frac{2}{\bar{X}^2} + \sum_{i=1}^{n}\frac{1}{X_i} &= \frac{1}{\bar{X}^2}\sum_{i=1}^{n}(X_i - 2\bar{X}) + \sum_{i=1}^{n}\frac{1}{X_i} && (63)\\
&= \frac{1}{\bar{X}^2}\sum_{i=1}^{n}(X_i - \bar{X}) - \frac{1}{\bar{X}^2}\sum_{i=1}^{n}\bar{X} + \sum_{i=1}^{n}\frac{1}{X_i} && (64)\\
&= 0 - \frac{1}{\bar{X}^2}\sum_{i=1}^{n}\bar{X} + \sum_{i=1}^{n}\frac{1}{X_i} && (65)\\
&= \sum_{i=1}^{n}\left(\frac{1}{X_i} - \frac{1}{\bar{X}}\right). && (66)
\end{aligned}$$

Thus,

$$\hat{\lambda}_n = \frac{n}{\sum\limits_{i=1}^{n} \left(\frac{1}{X_i} - \frac{1}{\bar{X}} \right)}.$$

$\blacksquare$